

ABSENCE OF BOUNDED-ORBIT WANDERING DOMAINS

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ABSTRACT. We prove that transcendental entire functions do not have bounded-orbit wandering domains. This answers a long-standing open question. In fact, we prove a more general theorem, establishing the absence of certain wandering domains even for locally defined analytic functions.

The proof is based on a recent remarkable argument of Ye, who obtained an elementary proof of Sullivan’s classical theorem that rational maps do not have wandering domains.

1. INTRODUCTION

Let $f: X \rightarrow X$ be either a rational function $f: \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$, or a transcendental entire function $f: \mathbb{C} \rightarrow \mathbb{C}$. We may consider f as the transition rule of a discrete-time dynamical system. The state space X then splits into two fundamental parts, defined and studied independently by Fatou and Julia in the early 20th century in the case of rational functions, and extended to transcendental entire functions by Fatou in 1926 [Fat26]. These sets are now called the *Fatou set* and the *Julia set*. The former is an open set and consists of those points where the dynamics of f is *stable* under perturbation. More formally, the iterates $\{f^n\}_{n=0}^\infty$ are equicontinuous (with respect to the spherical metric) near points of $F(f)$. The complement is the *Julia set* $J(f)$, where the dynamics is unstable. We refer to [Mil06] for background in holomorphic dynamics, particularly the iteration of rational functions, and to [Ber93] for a survey on the iteration of transcendental entire and meromorphic functions.

The connected components of $F(f)$ are called *Fatou components*. If U is a Fatou component, then for every $n \geq 0$, the n -th forward image $f^n(U)$ is contained in a Fatou component U_n . If $U_n \neq U_m$ for all $n \neq m$, then U is called *wandering*; otherwise, U is called *eventually periodic*. Eventually periodic Fatou components can be classified into a small finite number of different types, each of whose dynamics is well-understood.

Entire functions may have wandering domains, while rational maps do not by a famous theorem of Sullivan [Sul85]. It is known that every wandering domain U must belong to one of three types:

- (1) in an *escaping* wandering domain, $|f^n(z)| \rightarrow \infty$ for every $z \in U$;
- (2) in an *oscillating* wandering domain, $\liminf_{n \rightarrow \infty} |f^n(z)| < \infty = \limsup_{n \rightarrow \infty} |f^n(z)|$ for every $z \in U$;
- (3) in a *bounded-orbit* wandering domain, $\limsup_{n \rightarrow \infty} |f^n(z)| < \infty$ for every $z \in U$.

The first example of a wandering domain, constructed by Baker [Bak76], was an escaping wandering domain. Eremenko and Lyubich [EL87] constructed the first known

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examples of oscillating wandering domains. Wandering domains have been a highly active area of research recently; see for example [BEF⁺22, Boc21, EGPS23, PSS24, MRW25, EMR26] and the references in these papers. The existence of bounded-orbit wandering domains, on the other hand, is a long-standing open question, versions of which appeared three times on Hayman's list of *Research Problems in Complex Analysis* [HL19], in 1984 and 1989 (twice). They are listed as Problems 2.67, 2.77 and 2.87 and attributed to Baker, Eremenko, Herman, Kra and Lyubich.

Question 1.1 (Bounded-orbit wandering domains). *Does there exist a transcendental entire function with a bounded-orbit wandering domain?*

Orbits in hypothetical wandering domains of rational functions would automatically be “bounded” in the sense that they are compactly contained in the domain of definition (which is itself compact). Moreover, on a compact subset of the complex plane, there is no discernible difference between a polynomial, a transcendental entire function, and a general holomorphic function. Indeed, any holomorphic function on a neighbourhood of a compact set that does not separate the plane can be arbitrarily closely approximated by a polynomial, by Runge's theorem [Run85]. So it is natural to ask whether different bounded-orbit dynamics can arise for transcendental entire functions than for polynomials, or not. However, Sullivan's proof of the no wandering domains theorem uses deformation theory and relies crucially on the fact that rational maps of a given degree depend only on finitely many parameters. So it remained unclear whether one should expect the existence of bounded-orbit wandering domains for entire functions.

Recently, a remarkable new proof of Sullivan's theorem was proposed by Ye [Ye26]. This proof is essentially elementary, using only well-known properties of the hyperbolic metric in plane domains and the Gauss-Bonnet theorem. This is a truly remarkable development, given how long Sullivan's proof stood essentially without improvement – indeed, in 2014, the second author expressed strong scepticism that a simple proof of the no wandering domains theorem could be found [Rem]. The proof in [Ye26] also applies to transcendental entire functions with finitely many critical and asymptotic values, a result previously proved in [GK86] and [EL92] using Sullivan's techniques, and indeed to a more general class for which the theorem had not previously been established: those for which the derived set of the set of singular values is contained in the Fatou set.

In this paper, we further develop the ideas pioneered in [Ye26] to resolve Question 1.1 in the negative.

Theorem 1.2 (No bounded-orbit wandering domains). *Let f be a transcendental entire function, and let U be a wandering Fatou component of f . Then there exists a strictly increasing sequence $(n_k)_{k=0}^\infty$ such that $f^{n_k}|_U \rightarrow \infty$ locally uniformly as $k \rightarrow \infty$.*

In fact, we prove a more general statement for locally defined mappings, as follows. Suppose that $f: O \rightarrow \widehat{\mathbb{C}}$ is an open meromorphic function, where $O \subseteq \widehat{\mathbb{C}}$ is open. Given an open set $V \subseteq O$, we define the set

$$T_V(f) := \{z \in V : f^n(z) \in V \text{ for all } n \geq 0\} \quad (1.1)$$

of points that stay in V under iteration of f . Its interior

$$\Omega_V(f) := \text{int}(T_V) \quad (1.2)$$

consists of all points that have an entire neighbourhood that remains in V under iteration. Let U be a connected component of $\Omega_V(f)$. We say that U is *wandering* if $f^n(U)$ and $f^m(U)$ are contained in different connected components of $\Omega_V(f)$ for all $n \neq m$.

Theorem 1.3 (Local formulation on the sphere). *Let $O \subseteq \widehat{\mathbb{C}}$ be open and $f : O \rightarrow \widehat{\mathbb{C}}$ open and holomorphic. Let $V \Subset O$ be open with $\widehat{\mathbb{C}} \setminus \overline{V} \neq \emptyset$.*

- (1) *Let U be a wandering component of $\Omega_V(f)$, and let $z \in U$. Suppose that the connected component U_n of $\Omega_V(f)$ containing $f^n(U)$ is simply connected for every $n \geq 0$. Then there is a strictly increasing sequence $(n_k)_{k=0}^\infty$ such that $\text{dist}^\#(f^{n_k}(z), \partial V) \rightarrow 0$, where $\text{dist}^\#$ denotes spherical distance in $\widehat{\mathbb{C}}$.*
- (2) *Suppose that $A \subseteq T_V(f) \setminus \Omega_V(f)$ is measurable such that*
 - (a) *$f^n(A) \cap f^m(A) = \emptyset$ for $n \neq m$,*
 - (b) *f is injective on $\bigcup_{n=0}^\infty f^n(A)$, and*
 - (c) *A has positive area.*

Then $\bigcup_{n=0}^\infty f^n(A)$ is not contained in any compact subset of V .

The idea of the proof in [Ye26] relies on the fact that removing N points from a finitely punctured sphere increases the total hyperbolic area by $2\pi N$ (compare Lemma 2.5 below). This, together with the fact that f acts as a covering map over the complement of the set of singular values, is used to conclude that the hyperbolic area of the wandering domain in question is finite, which is a contradiction to the fact that the wandering domain is conformally equivalent to the hyperbolic plane, which has infinite area.

In our setting, we *cannot* exclude the singular values from our surfaces under consideration. The idea for dealing with this issue is as follows. If our wandering domain remains within a bounded part of the complex plane, then it is natural to remove only those critical values whose critical points likewise lie within this bounded part. The function is not a covering map when considering the preimage of the surface, but it behaves similarly to a covering on the part of the Riemann surface of the inverse function (thought of as spread over the plane) that corresponds to our bounded domain. The authors explored this idea in collaboration with generative AI, and arrived at the first version of the proof of Theorem 1.2. The proof presented here is fully written by the authors, and hopefully has added clarity and context over the version that first emerged from discussions with AI tools. The proof has also been auto-formalised in the Lean proof assistant using generative AI.

We note that we have formulated our proofs for subsets of the Riemann sphere, but they should carry over to functions $O \rightarrow \Sigma$, where Σ is a Riemann surface and $O \subseteq \Sigma$ is open. Using similar ideas, we have also been able to prove the following result, which answers another open question in transcendental dynamics: any wandering domain of a transcendental entire function f has a limit function that belongs to the derived set of singular values of f . The proof will be provided in a subsequent revision of the article.

Authors' statement on the use of generative AI. As mentioned above, generative AI (ChatGPT 6 Astra, 5.6 Sol, and Claude Opus 5) was used in a substantial way to develop the first draft of the proof of our main theorem, starting from our understanding of the proof in [Ye26] and our ideas for extending it. The proof was then presented and written by the authors in their own words. Generative AI (ChatGPT 6 Astra) was

also used to auto-formalise the proof in the Lean proof assistant, and to proof-read a completed draft of the paper.

Acknowledgements. Our rapid development of the ideas that led to the proof of Theorem 1.2 would not have been possible without the assistance of generative AI, nor without the key ideas in [Ye26], which according to the author’s statement was also assisted by AI. In turn, this assistance would undoubtedly not have been possible without the human mathematicians and thinkers who came before, and whose work is the basis on which modern AI tools have been built. It is our hope that the benefits of today’s powerful new AI tools will likewise be used to benefit mathematics, and humanity, as a whole.

2. PRELIMINARIES

Basic notation. As usual, we define the complex plane by $\mathbb{C}$ and the Riemann sphere by $\hat{\mathbb{C}}$. Boundaries and closures are taken in $\mathbb{C}$ if not explicitly stated otherwise. We write $V \Subset U$ to mean that V is *compactly contained* in U ; that is, the closure of V in U is a compact subset of U . Euclidean distance is denoted by dist , and Euclidean balls with centre z and radius r are denoted $B(z, r)$. If $x \in \mathbb{R}$, we set $x_+ := \max(x, 0)$.

Hyperbolic geometry. Suppose that X is a hyperbolic Riemann surface – that is, one which is not a torus, a sphere or a once- or twice-punctured sphere. Then X carries a unique complete conformal metric of constant curvature -1 , called the *hyperbolic metric*. In local coordinates, we may write this metric as $\rho_X(z)|dz|$, where ρ_X is a real-analytic function of z . In particular, if $X \subseteq \mathbb{C}$, then we may describe the metric globally by its *density* ρ_X with respect to the Euclidean metric. The condition that the metric has curvature -1 means that $\Delta \log \rho_X = \rho_X^2$. Hyperbolic distance in X is denoted dist_X , and hyperbolic discs are denoted $B_X(z, r)$.

The density of the hyperbolic metric on the unit disc is given by

$$\rho_{\mathbb{D}}(z) = \frac{2}{1 - |z|^2}.$$

A key fact is that holomorphic maps between hyperbolic surfaces do not expand hyperbolic metrics; see [Mil06, Theorem 2.11].

Theorem 2.1 (Pick’s theorem). *Let X and Y be hyperbolic surfaces, and let $f: X \rightarrow Y$ be holomorphic. Then the hyperbolic distortion of f at a point $z \in X$ is the quantity*

$$\|Df(z)\|_X^Y := |f'(z)| \cdot \frac{\rho_Y(f(z))}{\rho_X(z)}$$

(where the densities are understood in local coordinates).

This quantity satisfies

$$\|Df(z)\|_X^Y \leq 1$$

for all $z \in X$, with equality if and only if $f: X \rightarrow Y$ is a holomorphic covering map.

In particular, if $X \subseteq Y$, then $\rho_X \geq \rho_Y$.

We also make use of the following version of a standard hyperbolic density estimate; see [MR13, Proposition 3.4] or [Min17, Theorem 1].

Lemma 2.2. *There exist strictly decreasing, surjective and continuous functions*

$$\alpha, \beta: (0, \infty) \rightarrow (1, \infty)$$

with the following property. Let X be a hyperbolic Riemann surface and let $Y \subsetneq X$ be open and $z \in Y$. Then

$$1 \leq \alpha(\delta) \leq \frac{\rho_Y(z)}{\rho_X(z)} \leq \beta(\delta),$$

where $\delta = \text{dist}_X(z, X \setminus Y)$, and the quotient of densities should be understood in local coordinates.

Remark 2.3. The exact expressions for α and β are given in [MR13, Proposition 3.4], but we do not require them.

In our proofs, we will approximate hyperbolic domains of the plane by finitely punctured spheres. The relation between the hyperbolic metric of the limit to that of its approximations is described by the following qualitative lemma. (See also [Ye26, Lemma 2.2].)

Lemma 2.4. *Let $J \subseteq \mathbb{C}$ be closed and non-empty, and let $(P_j)_{j=0}^\infty$ be an increasing sequence of finite subsets with $\#P_j \geq 2$, and such that $P_\infty := \bigcup_{j=0}^\infty P_j$ is dense in J . Then:*

- (1) *If $z \in J \setminus P_\infty$, then $\rho_{\mathbb{C} \setminus P_j}(z) \rightarrow \infty$ as $j \rightarrow \infty$.*
- (2) *If z belongs to a connected component U of $\mathbb{C} \setminus J$, then $\rho_{\mathbb{C} \setminus P_j}(z)$ converges locally uniformly on U to $\rho_U(z)$ as $j \rightarrow \infty$.*

Proof. Let $z \in J \setminus P_\infty$. Then $\lim_{j \rightarrow \infty} \text{dist}(z, P_j) = 0$ by choice of the sets P_j . The hyperbolic metric of $\mathbb{C} \setminus P_0$ and the Euclidean metric are comparable in a neighbourhood of z , so also $\lim_{j \rightarrow \infty} \text{dist}_{\mathbb{C} \setminus P_0}(z, P_j) = 0$. By Lemma 2.2, $\rho_{\mathbb{C} \setminus P_j}(z)/\rho_{\mathbb{C} \setminus P_0}(z) \rightarrow \infty$, establishing (1).

To prove 2, by Lemma 2.2 it is enough to show that

$$\lim_{j \rightarrow \infty} \text{dist}_{\mathbb{C} \setminus P_j}(z, \zeta) \rightarrow \infty$$

for every $z \in U$ and every $\zeta \in \partial U \setminus P_\infty$. To see this, let $\varepsilon > 0$ and let n be so large that there are $p_1, p_2 \in P_n$ with $\text{dist}(\zeta, p_2) < \text{dist}(p_1, p_2) < \varepsilon$. Let $\lambda: \mathbb{C} \rightarrow \mathbb{C}$ be the affine map that takes p_1 to 0 and p_2 to 1; then $\lambda(\zeta)$ remains bounded as $\varepsilon \rightarrow 0$, while $\lambda(z) \rightarrow \infty$. We conclude that

$$\text{dist}_{\mathbb{C} \setminus P_j}(z, \zeta) \geq \text{dist}_{\mathbb{C} \setminus \{p_1, p_2\}}(z, \zeta) = \text{dist}_{\mathbb{C} \setminus \{0, 1\}}(\lambda(z), \lambda(\zeta)) \rightarrow \infty$$

as $\varepsilon \rightarrow 0$. ■

Lemma 2.5. *Let $P, Q \subseteq \widehat{\mathbb{C}}$ be two finite subsets such that $P \subseteq Q$ and $|P| \geq 3$. Then for any measurable subset $U \subseteq \widehat{\mathbb{C}} \setminus Q$,*

$$\text{Area}_{\widehat{\mathbb{C}} \setminus Q}(U) - \text{Area}_{\widehat{\mathbb{C}} \setminus P}(U) \leq 2\pi(|Q| - |P|).$$

Proof. It is a classical consequence of the Gauss–Bonnet theorem that the hyperbolic area of an n -times punctured sphere is $2\pi(n - 2)$; see [Bea83, Theorem 10.4.3]. The claim follows from Pick’s theorem. ■

Subharmonic functions. We use the following standard facts concerning subharmonic functions. Suppose $D \subseteq \mathbb{C}$ is a domain, then a function $u: D \rightarrow [-\infty, \infty)$ is subharmonic if it is upper semicontinuous and satisfies the mean value inequality

$$u(z) \leq \frac{1}{2\pi} \int_0^{2\pi} u(z + re^{it}) dt,$$

for each $\overline{B(z, r)} \subseteq D$. If $u \in C^2(D)$, then this is equivalent to $\Delta u \geq 0$, where Δ denotes the Laplacian.

If u is subharmonic, the distributional Laplacian Δu is positive in the sense of distributions. Hence by the Riesz representation theorem there exists a unique positive measure μ_u on D such that

$$\int_D \chi d\mu_u = \int_D u \Delta \chi dA$$

for every $\chi \in C_c^\infty(D)$. We call μ_u the Riesz measure of u . See [HK76].

3. UNIFORM AREA ESTIMATES ON COMPACT SETS

In this section, we prove the key result for our proof, which concerns a comparison of hyperbolic areas. In order to explain the idea, we briefly review Ye's proof of Sullivan's theorem. Suppose that f is a rational function. We may choose some finite forward-invariant approximation P of the Julia set; that is, $P \subseteq J(f)$ and every point of $J(f)$ is within $\varepsilon > 0$ of some point in P . Consider the finitely punctured spheres $X := \widehat{\mathbb{C}} \setminus P$ and $Y := X \setminus S(f)$, where $S(f)$ is the set of critical values of f . We have $f^{-1}(Y) \subseteq X$ and $f: f^{-1}(Y) \rightarrow Y$ is a covering map. So by Pick's theorem, the hyperbolic metric on $f^{-1}(Y)$, which is precisely the pull-back of the hyperbolic metric on Y under f , is no smaller than that on X .

Now suppose that $(U_n)_{n=0}^\infty$ is an orbit of simply connected wandering domains such that the map f is univalent along the orbit U_n . (A well-known argument shows that some forward image of a wandering domain of a rational function would have this property.) Then the above argument shows that the hyperbolic area of the wandering domains does not decrease under f , considered as a map from a subset of X to Y , so

$$\text{Area}_Y \left(\bigcup_{j=1}^\infty U_n \right) \geq \text{Area}_X \left(\bigcup_{j=0}^\infty U_n \right).$$

But only U_0 is missing from the upper union compared to the lower one, and removing the finitely many points of $S(f)$ from X increases the total hyperbolic area by at most $2\pi \#S(f)$. So we see that $\text{Area}_X(U_0)$ is bounded by $2\pi \#S(f)$, which is a bound that is *independent* of the size of P . On the other hand, if we choose P sufficiently large, then the hyperbolic metric of X on U converges to the hyperbolic metric on U , which is a simply connected domain and therefore has infinite hyperbolic area. This yields the desired contradiction.

The main difficulty that we need to overcome is that, for a general transcendental entire function, the set $S(f)$ of singular values may be infinite; indeed, it may coincide with the entire complex plane. The key observation is the following. Form X as before, but restrict our attention to a compact part K of the complex plane. Let V be an open

neighbourhood of K and, in defining the surface Y , only remove those critical values that correspond to critical points in V . Then $f: f^{-1}(Y) \rightarrow Y$ is no longer a covering, so if we pass from the hyperbolic metric of X to the pull-back η_P of the hyperbolic metric of Y , the metric may decrease in some places. The goal of this section is to show that the total decrease of such area is bounded by a constant, independently of P . This is the statement of the following theorem, which allows the remainder of the argument to go through similarly as before.

Theorem 3.1 (Comparison of hyperbolic area and pullback). *Let $V \subseteq \mathbb{C}$ be bounded and open, and let f be open and meromorphic on a neighbourhood of $\overline{V}$. Fix a compact set $K \subseteq V$ and distinct $a, b \in \mathbb{C}$. Then there is $C \geq 0$ with the following property.*

Define $E := f(\{z \in V : f'(z) = 0\})$. For every finite set $P \supset \{a, b\}$ satisfying $f(P \cap V) \subseteq P$, write

$$X_P := \mathbb{C} \setminus P, \quad Y_P := \mathbb{C} \setminus (P \cup E)$$

and let η_P be the density of the pullback of the hyperbolic metric of Y_P under f :

$$\eta_P(z) := |f'(z)|\rho_{Y_P}(f(z)), \quad z \in f^{-1}(Y_P).$$

Then

$$\int_{K \cap f^{-1}(Y_P)} (\rho_{X_P}^2 - \eta_P^2)_+ dA \leq C.$$

To prove Theorem 3.1, we first bound the ratio of the corresponding hyperbolic metrics by a uniform constant.

Lemma 3.2 (Comparison of hyperbolic metric and pullback). *Under the assumptions of Theorem 3.1, there is a constant $M \geq 1$, which is independent of P , such that*

$$\frac{\rho_{X_P}(z)}{\eta_P(z)} \leq M$$

for all $z \in K \cap f^{-1}(Y_P)$.

Proof. Observe that $\infty \notin Y_P$, so f has no poles in $f^{-1}(Y_P)$, so the density $\eta_P(z)$ is indeed defined and finite.

Let $z_0 \in K$ and choose neighbourhoods $G' \Subset G \Subset V$ of z_0 and a disc D centred at $f(z_0)$ so that $f: G \rightarrow D$ is proper, and $f(\overline{G'}) \Subset D$. (If z_0 is a pole, then D is chosen to be a disc on the Riemann sphere centred at ∞ , and the calculations below should be understood to be carried out in local coordinates.)

Now, note that the restriction $f: G \cap f^{-1}(Y_P) \rightarrow D \cap Y_P$ is a covering map and $G \cap f^{-1}(Y_P) \subseteq X_P$. Indeed, if $z \in G \cap f^{-1}(Y_P)$ were in P , then $f(z) \in P$ by $f(P \cap V) \subseteq P$, contradicting $f(z) \in Y_P$. It follows by Pick's theorem that

$$\rho_{X_P}(z) \leq \rho_{G \cap f^{-1}(Y_P)}(z) = |f'(z)|\rho_{D \cap Y_P}(f(z)).$$

Hence,

$$\frac{\rho_{X_P}(z)}{\eta_P(z)} \leq \frac{\rho_{D \cap Y_P}(f(z))}{\rho_{Y_P}(f(z))},$$

for $z \in G' \cap f^{-1}(Y_P)$. It remains to compare the hyperbolic metrics of $D \cap Y_P$ and Y_P on $f(G')$.

Now, since $a, b \in P$, by the definition of Y_P we have that $\rho_{Y_P} \geq \rho_{\mathbb{C} \setminus \{a, b\}}$. Moreover, the image of $\overline{G'}$ is compactly contained in D . The density $\rho_{\mathbb{C} \setminus \{a, b\}}$ is bounded from below on $D \setminus \{a, b\}$ and the Euclidean distance between $f(\overline{G'})$ and ∂D is positive. So there exists $\delta > 0$, independent of P , such that $d_{Y_P}(z, Y_P \setminus D) \geq \delta$ for $z \in f(G') \cap Y_P$. By Lemma 2.2, we obtain a constant $M(G')$, independent of P , so that

$$\frac{\rho_{D \cap Y_P}(f(z))}{\rho_{Y_P}(f(z))} \leq M(G'),$$

for $z \in f(G') \cap Y_P$. Finally, finitely many neighbourhoods G' cover K , so we may take the maximum of each of the corresponding finitely many constants $M(G')$ to obtain M . $\blacksquare$

The idea now is that, in the situation of Lemma 3.2, the function $\log \left(\frac{\rho_{X_P}}{\eta_P} \right)_+$ is a subharmonic function on $V \cap f^{-1}(Y_P)$, which is locally bounded near each point of $V \setminus f^{-1}(Y_P)$. As a result, we will be able to bound its integral on K by a bound that is independent of P , using the following fact about subharmonic functions.

Lemma 3.3. *Let F be a finite subset of an open set $V \subseteq \mathbb{C}$, and let $u \in C^2(V \setminus F)$ be locally bounded above near F . Suppose that $u(z) > 0$ if and only if $\Delta u(z) > 0$, for all $z \in V \setminus F$. Let $\chi \in C_c^\infty(V)$ be a non-negative test function, and suppose that there are $m \geq 0$ and a neighbourhood U of $\text{supp } \Delta \chi \subseteq \text{supp } \chi$ such that $u(z) \leq m$ for all $z \in U \setminus F$. Then*

$$\int_{V \setminus F} \chi \cdot (\Delta u)_+ \, dA \leq m \int_V |\Delta \chi| \, dA.$$

Proof. Set $v = u_+$ on $V \setminus F$. Then v is subharmonic on $V \setminus F$, since on $\{z : u(z) > 0\}$ we have $v = u$ and $\Delta u > 0$. On the other hand, if $u(z) \leq 0$, then $v(z) = 0$, and v satisfies the mean value inequality at z , since v is globally nonnegative.

Now u is locally bounded above near F , and hence so is v . Because F is a finite (and so polar) set, [HK76, Theorem 5.18] implies that v extends to a subharmonic function on V , with $v(z) \leq m$ for $z \in U$.

Consider the Riesz measure of v , which we denote μ_v . On $\{z : u(z) > 0\}$, we have $v = u$ by definition, so $\mu_v = \Delta u \, dA$. Also, since $u(z) > 0$ if and only if $\Delta u > 0$, we have that $(\Delta u)_+ = 0$ on $\{z : u(z) \leq 0\}$. Hence,

$$\mu_v \geq 1_{V \setminus F} \cdot (\Delta u)_+ \, dA.$$

Since χ is nonnegative,

$$\int_{V \setminus F} \chi \cdot (\Delta u)_+ \, dA \leq \int_V \chi \, d\mu_v.$$

Now by the definition of the Riesz measure,

$$\int_V \chi \, d\mu_v = \int_V v \Delta \chi \, dA.$$

Finally, since $v \leq m$ on U , which is a neighbourhood of the support of $\Delta \chi$,

$$\int_V v \Delta \chi \, dA \leq m \int_V |\Delta \chi| \, dA.$$

Combining the inequalities, we obtain the desired result. $\blacksquare$

Proof of Theorem 3.1. Choose a compact set $K' \subseteq V$ with $K \subseteq \text{int}(K')$ and choose a nonnegative test function $\chi \in C_c^\infty(K')$ such that $\chi = 1$ on a neighbourhood of K . By Lemma 3.2 applied to K' , there is $M \geq 1$, independent of P , such that

$$\frac{\rho_{X_P}(z)}{\eta_P(z)} \leq M$$

for all $z \in K' \cap f^{-1}(Y_P)$.

Let $F := V \setminus f^{-1}(Y_P) = V \cap f^{-1}(P \cup E \cup \{\infty\})$, which is a finite set, and define $u(z) := \log(\rho_{X_P}(z)/\eta_P(z))$ on $V \setminus F$. By Lemma 3.2, u is locally bounded above near F . In particular, $u \leq \log M$ on a neighbourhood of the support of $\Delta\chi$, since the support of $\Delta\chi$ is compactly contained in K' .

Since hyperbolic metrics have curvature -1 , $\Delta \log \rho_{X_P} = \rho_{X_P}^2$, and similarly for ρ_{Y_P} . Since curvature is a conformal invariant and $f'(z) \neq 0$ on $V \setminus F$, the pullback η_P of ρ_{Y_P} under f also has constant curvature -1 , i.e. $\Delta \log \eta_P = \eta_P^2$. (This can also be checked directly by expanding

$$\Delta \log \eta_P = \Delta \log |f'| + \Delta(\log \rho_{Y_P} \circ f) = |f'|^2(\Delta \log \rho_{Y_P}) \circ f,$$

and then substituting $\Delta \log \rho_{Y_P} = \rho_{Y_P}^2$.)

Hence, our function u satisfies $\Delta u = \rho_{X_P}^2 - \eta_P^2$. Moreover, $u(z) > 0$ if and only if $\rho_{X_P}(z) > \eta_P(z)$ if and only if $\Delta u(z) > 0$. We can thus apply Lemma 3.3 to u , which gives

$$\int_{V \setminus F} \chi \cdot (\rho_{X_P}^2 - \eta_P^2)_+ \, dA \leq \log M \int_V |\Delta \chi| \, dA,$$

and since $\chi = 1$ on K ,

$$\int_{K \cap f^{-1}(Y_P)} (\rho_{X_P}^2 - \eta_P^2)_+ \, dA \leq \log M \int_V |\Delta \chi| \, dA =: C \geq 0,$$

as claimed. (Note that the right hand side is indeed independent of P .) ■

4. ABSENCE OF BOUNDED-ORBIT WANDERING DOMAINS FOR LOCALLY DEFINED FUNCTIONS

With Theorem 3.1 in place, we are now ready to carry the central argument of [Ye26] over to our more general setting, where it takes the following form.

Lemma 4.1 (Uniform area bound). *We continue to use the notation of Theorem 3.1, with its fixed compact set K and constant C . Let $W \subseteq K \cap f^{-1}(Y_P)$ be measurable and suppose that $f|_W$ is injective. If $B \subseteq W$ is measurable and $f(W) \subseteq W \setminus B$, then*

$$\text{Area}_{X_P}(B) \leq C + 2\pi|E|.$$

Proof. Theorem 3.1 implies that

$$\begin{aligned} \text{Area}_{X_P}(W) &= \int_W \rho_{X_P}^2 \, dA \leq \int_W \eta_P^2 \, dA + \int_W (\rho_{X_P}^2 - \eta_P^2)_+ \, dA \leq \int_W \eta_P^2 \, dA + C \\ &= \int_W |f'(z)|^2 \rho_{Y_P}^2(f(z)) \, dA + C = \text{Area}_{Y_P}(f(W)) + C. \end{aligned}$$

On the other hand, by Lemma 2.5, $\text{Area}_{Y_P}(f(W)) \leq \text{Area}_{X_P}(f(W)) + 2\pi|E|$. So

$$\begin{aligned} \text{Area}_{X_P}(W) &\leq \text{Area}_{Y_P}(f(W)) + C \leq \text{Area}_{X_P}(f(W)) + C + 2\pi|E| \\ &\leq \text{Area}_{X_P}(W \setminus B) + C + 2\pi|E|. \end{aligned}$$

Substituting $\text{Area}_{X_P}(W) = \text{Area}_{X_P}(B) + \text{Area}_{X_P}(W \setminus B)$ gives the desired inequality. $\blacksquare$

For locally defined functions, the finite invariant sets P_j that we wish to approximate the “Julia set” can no longer necessarily be chosen to be periodic points. Instead, we will obtain them by taking preimages of finite subsets of the boundary of their domain of definition.

Lemma 4.2. *Let $V \subseteq \mathbb{C}$ be a bounded open set, and let $f: V \rightarrow \widehat{\mathbb{C}}$ be meromorphic on a neighbourhood of $\overline{V}$ and non-constant on every component of V . Then there exists an increasing sequence of finite subsets $(P_j)_{j=0}^\infty$ of $\overline{V}$ such that the following hold.*

- (1) $|P_j| \geq 2$ for all $j \geq 0$.
- (2) $f(P_j \cap V) \subseteq P_j$ for all $j \geq 0$.
- (3) $J \cap \Omega_V(f) = \emptyset$, where $J := \overline{\bigcup_{j=1}^\infty P_j}$.
- (4) For every connected component U of $\mathbb{C} \setminus J$, either U is a connected component of $\Omega_V(f)$, or $U \cap T_V(f) = \emptyset$.

Proof. Let $(Q_j)_{j=0}^\infty$ be an increasing sequence of finite subsets of ∂V such that $|Q_0| \geq 2$ and $\overline{\bigcup_{j=1}^\infty Q_j} = \partial V$. Define

$$T_{j,0} := Q_j, \quad T_{j,r+1} := T_{j,r} \cup (V \cap f^{-1}(T_{j,r})), \quad \text{and} \quad P_j = T_{j,j}.$$

Clearly, (P_j) forms an increasing sequence of finite sets satisfying (1) and (2).

Set $P_\infty := \bigcup_{j=1}^\infty P_j$. Since $\bigcup_{j=1}^\infty Q_j$ is dense in ∂V , we have $\partial V \subseteq J$. Moreover, every point of $P_\infty \cap V$ eventually maps to ∂V , and hence does not belong to $T_V(f)$. Since $\Omega_V(f)$ is open, it follows that $J \cap \Omega_V(f) = \emptyset$, proving (3).

We next observe that $V \cap f^{-1}(J) \subseteq J$. Indeed, suppose that $z \in V$ and $f(z) \in J$. Let $W \Subset V$ be any neighbourhood of z . Then $f(W)$ is a neighbourhood of $f(z)$, and hence contains some point $a \in P_\infty$, say $a \in P_j$. Choose $w \in W$ with $f(w) = a$. Since $P_j = T_{j,j}$ and $T_{j,j} \subseteq T_{j+1,j}$, we have $w \in T_{j+1,j+1} = P_{j+1}$. Thus every neighbourhood of z intersects P_∞ , so $z \in J$.

Now let U be a component of $\mathbb{C} \setminus J$ such that $U \cap T_V(f) \neq \emptyset$, and choose $z \in U \cap T_V(f)$. Since $\partial V \subseteq J$ and U is connected, we have $U \subseteq V$. The previously proved inclusion $V \cap f^{-1}(J) \subseteq J$ also implies that $f(U) \cap J = \emptyset$. Since $f(U)$ is connected and contains $f(z) \in V$, it is contained in a connected component U_1 of $\mathbb{C} \setminus J$ with $U_1 \cap T_V(f) \neq \emptyset$. Inductively, we see that $f^n(U) \subseteq V$ for every $n \geq 0$. Hence $U \subseteq T_V(f)$, and since U is open, $U \subseteq \Omega_V(f)$. Moreover, $\partial U \subseteq J$ and $J \cap \Omega_V(f) = \emptyset$, so U is indeed a connected component of $\Omega_V(f)$. This proves (4). $\blacksquare$

Lemma 4.3. *Let $O \subseteq \widehat{\mathbb{C}}$ be open and $f: O \rightarrow \widehat{\mathbb{C}}$ open and meromorphic. Let $V \Subset O$ be a bounded open subset of $\mathbb{C}$. Let U be a wandering component of $\Omega_V(f)$ such that the connected component U_n of $\Omega_V(f)$ containing $f^n(U)$ is simply connected for every $n \geq 0$. For $R > 0$, let $B_{U_n}(z_n, R)$ denote the hyperbolic ball in U_n of radius R centered at z_n . Then*

- (1) the Euclidean diameter of $B_{U_n}(z_n, R)$ tends to 0 as $n \rightarrow \infty$;
- (2) if the closure of the orbit $(z_n)_{n=0}^\infty$ is compactly contained in V , then there exists $N = N(R)$ such that, for every $n \geq N(R)$, the restriction $f|_{B_{U_n}(z_n, R)}$ is univalent.

Proof. Let $\pi_n: \mathbb{D} \rightarrow U_n$ be a Riemann map such that $\pi_n(0) = z_n$. Observe that $B_{U_n}(z_n, R) = \pi_n(B_{\mathbb{D}}(0, r))$ for some $r \in (0, 1)$ depending only on R .

Suppose that $\text{dist}(z_n, \partial U_n)$ does not tend to 0. Then there exist $\delta > 0$ and infinitely many n such that $\mathbb{D}_\delta(z_n) \subseteq U_n$. This is impossible since the domains U_n are pairwise disjoint and the points z_n remain in the fixed bounded set V .

Hence $\text{dist}(z_n, \partial U_n) \rightarrow 0$. By Koebe's 1/4-theorem,

$$|\pi'_n(0)| \leq 4 \text{dist}(\pi_n(0), \partial U_n) \rightarrow 0.$$

At the same time, by Koebe's distortion theorem,

$$|\pi_n(w) - \pi_n(0)| \leq \frac{r}{(1-r)^2} |\pi'_n(0)|,$$

for $|w| \leq r$. It follows that the diameter of $B_{U_n}(z_n, R)$ tends to zero.

Now let us establish (2). Let K be the closure of the orbit (z_n) ; then K is forward-invariant and, in particular, contains no poles of f . Choose finitely many sufficiently small Jordan domains $D_i \Subset V$ and Jordan domains $D'_i \Subset D_i$ so that the D'_i cover K and

$$f(z) = (\varphi_i \circ g_d \circ \psi_i^{-1})(z) \quad \text{for } z \in D_i,$$

where $g_d(z) = z^d$ for $d \geq 1$, and $\varphi_i: \mathbb{D} \rightarrow f(D_i) =: G_i$ and $\psi_i: \mathbb{D} \rightarrow D_i$ are biholomorphisms such that $\varphi_i(0) = f(w_i)$ and $\psi_i(0) = w_i$. In particular, f has no critical points in D_i except possibly w_i .

Since the diameter of $B_n(R)$ tends to 0, we may assume that, whenever $z_n \in D'_i$, we have $B_n(R) \Subset D_i$. Also, because the set $E := f(\{z \in V : f'(z) = 0\})$ of critical values of f in V is finite and our domain is wandering, for all sufficiently large n , U_n , and hence $B_n(R)$, is disjoint from E .

Again, since the diameter of $B_n(R)$ tends to 0, we may assume that $\overline{B_{n+1}(R)} \subseteq G_i$, and as above we may assume that $B_{n+1}(R) \cap E = \emptyset$. It then follows from the definition of G_i that each component of $f^{-1}(B_{n+1}(R)) \cap D_i$ is mapped univalently onto $B_{n+1}(R)$. By Pick's theorem, $f(B_n(R)) \subseteq B_{n+1}(R)$, and hence $B_n(R)$ is contained in the component containing z_n . Therefore, $f|_{B_n(R)}$ is univalent, as required. $\blacksquare$

Now we are ready to prove Theorem 1.3.

Proof of Theorem 1.3(1). After conjugating f by a Möbius transformation, we may assume that V is bounded in $\mathbb{C}$. Suppose that the orbit $(z_n)_{n=0}^\infty$ is compactly contained in V . Choose a compact set $K \subseteq V$ such that the closure of the orbit is compactly contained in $\text{int}(K)$. We also choose sets P_j as in Lemma 4.2 and distinct points $a, b \in P_1$. Let the finite set E and the constant C be chosen as in Theorem 3.1.

Fix $R > 0$. By Lemma 4.3, $B_n(R) := B_{U_n}(z_n, R) \subseteq K$ and $f|_{B_n(R)}$ is injective for all $n \geq N(R)$. Since $f^{-1}(E) \cap V$ is finite, we may also assume that $U_n \cap f^{-1}(E) = \emptyset$ for all $n \geq N(R)$.

By Pick's theorem, we have $f(B_n(R)) \subseteq B_{n+1}(R)$. Define the union $W_R := \bigcup_{n=N(R)}^\infty B_n(R)$. By construction, $W_R \subseteq K \setminus f^{-1}(P_j \cup E)$ for all $j \geq 1$, and f maps W_R injectively into $W_R \setminus B_{N(R)}(R)$.

Lemma 4.1, applied with $P = P_j$, now gives $\text{Area}_{X_j}(B_{N(R)}(R)) \leq C + 2\pi|E|$ for every j , where $X_j = \mathbb{C} \setminus P_j$. By Lemma 2.4(2), ρ_{X_j} converges locally uniformly on $U_{N(R)}$ to $\rho_{U_{N(R)}}$. Hence

$$\text{Area}_{U_{N(R)}}(B_{N(R)}(R)) \leq C + 2\pi|E|,$$

with a bound independent of R . This leads to a contradiction, since

$$\text{Area}_{U_{N(R)}}(B_{N(R)}(R)) = \text{Area}_{\mathbb{D}}(B_{\mathbb{D}}(0, R))$$

tends to ∞ as $R \rightarrow \infty$, since the area of $\rho_{\mathbb{D}}$ is infinite. The proof is complete. $\blacksquare$

Proof of Theorem 1.3(2). Again, after conjugating f by a Möbius transformation, we may assume that V is bounded. Suppose there exists a measurable set $A \subseteq T_V \setminus \Omega_V$ such that the sets $f^n(A)$ are pairwise disjoint, f is injective on $W := \bigcup_{n=0}^{\infty} f^n(A)$, and W is compactly contained in V . We will show that A has area zero.

Let $K \subseteq V$ be a compact set such that $W \subseteq K$. We also choose sets P_j as in Lemma 4.2 and distinct points $a, b \in P_1$. Let the finite set E and the constant C be chosen as in Theorem 3.1.

Define $S_0 := P_{\infty} \cup E$, where $P_{\infty} = \bigcup_{j=1}^{\infty} P_j$. Inductively, define $S_{r+1} := S_r \cup (V \cap f^{-1}(S_r))$, and let $S_{\infty} := \bigcup_{r=0}^{\infty} S_r$. Set $A^* := A \setminus S_{\infty}$. Recall that $T_V \setminus \Omega_V \subseteq J = \overline{P_{\infty}}$ by Lemma 4.2 (4). So $A^* \subseteq J \setminus P_{\infty}$. Since S_{∞} is countable, removing it from A does not change its area. Let $W^* := \bigcup_{n=0}^{\infty} f^n(A^*)$. Clearly, $f|_{W^*}$ is injective, $f(W^*) = W^* \setminus A^*$, and $W^* \subseteq K \setminus f^{-1}(P_j \cup E)$ for all $j \geq 1$.

Lemma 4.1, applied with $P = P_j$, now gives $\text{Area}_{X_j}(A^*) \leq C + 2\pi|E|$ for every j , where $X_j = \mathbb{C} \setminus P_j$. It is also easy to see that $A^* \subseteq \overline{P_{\infty}}^{\mathbb{C}} \setminus P_{\infty}$. Since the densities $\rho_{X_j}(z)$ are increasing by Pick's theorem, the monotone convergence theorem gives

$$\lim_{j \rightarrow \infty} \text{Area}_{X_j}(A^*) = \int_{A^*} \lim_{j \rightarrow \infty} \rho_{X_j}^2 dA \leq C + 2\pi|E|.$$

However, by Lemma 2.4(1), $\rho_{X_j}(z)$ tends to ∞ as $j \rightarrow \infty$ for every $z \in A^*$. Therefore, A^* must have measure zero. $\blacksquare$

5. ABSENCE OF BOUNDED-ORBIT WANDERING DOMAINS FOR TRANSCENDENTAL ENTIRE FUNCTIONS

Proof of Theorem 1.2. Let f be a transcendental entire function with an orbit $(U_n)_{n=0}^{\infty}$ of wandering domains. Suppose, by way of contradiction, that the conclusion of the theorem does not hold; that is, U is a bounded-orbit wandering domain of f .

By definition of the Fatou set, $f^n|_U$ is a normal family in the sense of Montel (recall that normality is equivalent to equicontinuity by the Arzelà–Ascoli theorem). It is well-known that all limit functions of the family of iterates in a wandering domain are constant, and all finite such constants belong to the Julia set [Ber93, p. 18]. The assumption means that the collection of these constants is a compact set $L \subseteq \mathbb{C}$.

Let $z_0 \in U_0$ and let D be a disc centred at z_0 with $\overline{D} \subseteq U_0$. Let V be an open Euclidean disc centred at the origin, chosen large enough to ensure that $L \subseteq V$. Then, there is n_0 such that $f^n(\overline{D}) \subseteq V$ for $n \geq n_0$.

For $n \geq n_0$, let $\tilde{U}_n$ be the connected component of the set $\Omega_V(f)$ defined in (1.2) that contains $f^n(D)$. By the maximum principle, each $\tilde{U}_n$ is simply connected, and by Montel's theorem, $\tilde{U}_n \subseteq F(f)$, and hence $\tilde{U}_n \subseteq U_n$.

So $(\tilde{U}_n)_{n=n_0}^\infty$ is an orbit of simply connected wandering components of $\Omega_V(f)$, and hence satisfies the hypotheses of Theorem 1.3 (1). Hence there is a strictly increasing sequence $(m_k)_{k=0}^\infty$ such that $\text{dist}^\#(f^{n_0+m_k}(z), \partial V) \rightarrow 0$, which is a contradiction to the fact that every limit of the sequence $(f^n(z))_{n=0}^\infty$ belongs to the compact set $L \subseteq V$. ■

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